



FACULTY
OF EDUCATION

Masaryk University

Documentation (Presentation #5)

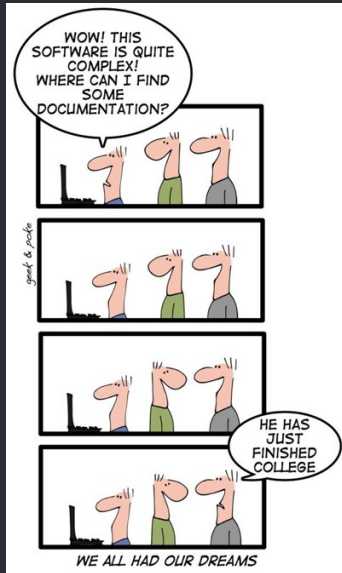
Special Topics (CS3001)

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A Brief Introduction

One of the most critical aspects of computer science is documentation.

A Brief Introduction



A Brief Introduction

- Documentation was one of those things discussed once briefly in Introduction to Programming. You were expected to do it, but never really taught how to do it.
- One of my classes this semester (Object Oriented Numerical Modeling) require *a lot* of documentation.
- Throughout the semester, I wanted to write good, thorough documentation.

Prior Knowledge

- Mediocre documentation is easy to write.
 - This is the kind of documentation I wrote my freshmen year.
- Great documentation is much harder to write.
 - This is documentation I strive to write now.

Tools

- Often, it's hard to read documentation when it'd directly inside code.
- This is why I've picked up Doxygen, a tool to scrap all documentation outside of a codebase.

Tools (Example)

Eigen
A Linear Algebra Library

Main Page Classes Files

Matrix<T> Class Template Reference [matrix](#)

Public Member Functions | List of all members

An interface for a Matrix. More...

#include <matrix.h>

Inheritance diagram for Matrix<T>:

```
graph TD; MatrixT["Matrix<T>"] --|> BaseMatrixT["BaseMatrix<T>"]; BaseMatrixT --|> BandedMatrixT["BandedMatrix<T>"]; BaseMatrixT --|> DiagonalMatrixT["DiagonalMatrix<T>"]; BaseMatrixT --|> RectangularMatrixT["RectangularMatrix<T>"]; BaseMatrixT --|> SymmetricMatrixT["SymmetricMatrix<T>"];
```

Public Member Functions

virtual void	resize	(const size_t rows, const size_t columns)=0
virtual T &	operator()	(const int row, const int column)=0
virtual const T &	operator()	(const int row, const int column) const =0
virtual T &	zero	() const noexcept=0
virtual size_t	rows	() const noexcept=0
virtual size_t	columns	() const noexcept=0

Detailed Description

template-class T>
class Matrix<T>

An interface for a Matrix.

Template Parameters

- T The type of the elements the matrix stores.

The documentation for this class was generated from the following file:

- libmatrix.h

Generated by [doxygen](#) 1.8.14

Goals

- To be able to use Doxygen efficiently.
- To write documentation that's easy to read, but also informative.

Before — Intro To Programming

```
// Description:      To get user input to dictate what menu option
                     is selected.
// Pre-Condition:    A valid character must be entered.
// Post-Condition:    A value for charMenuOption is received,
// dictating what functions are used form there.
```

After — Object Oriented Numerical Modeling

```
/**
 * @brief Read the specified input file and ensure proper
 *        data format.
 *
 * @param filename The filename where file is to be read from.
 *        Can be relative or absolute.
 *
 * @return An `std::string` with the entirety of the file
 *        contents in them (`\n` and all).
 *
 * @warning This does terminate prematurely if the file
 *        contents is not in the proper format.
 **/
```

Resources

- A great starting point was a [Fermilab introduction article](#).
- From there, I would reference:
 - [The doxygen website](#).
 - [The special doxygen command guide](#).
- I would also read articles about better documentation.
 - [HackerNoon](#).
 - [Documentation Handbook](#).
 - [Hacker News](#).

Goal Accomplishment

- So far, this semester, I have only missed two points on documentation for all of my classes.
 - Some of my graders even commented on my documentation ability (see attached).
- I have been more conscious about my technical writing.
- Overall, I saw an improvement in my documentation skills. This is reflected in my school work and my work life.

In Closing

All question, comments, and insults can be directed towards me:

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